CURRICULUM VITAE

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NAME: Okeh, Debra Ukamaka

POSITION TITLE: Programme Associate, Infectious Diseases

EDUCATION/TRAINING:

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
University of Nigeria	MBBS	03/2009	Medicine and Surgery
Brandeis University, United States	MS	05/2018	Global Health Policy and Management
University of Port Harcourt	MSc	2025	One Health & Disease Modelling

A. Personal Statement

I am a global health physician specialising in infectious disease epidemiology and applied mathematical modelling. Currently, I lead the team providing technical assistance to the National Malaria Elimination Programme (NMEP) for the planning and implementation of the Malaria Indicator Survey (MIS) 2025. This role requires working closely with NMEP and National Population Commission to leverage the structures for enumeration areas (clusters) and systems using CSPro for the MIS. We also support the planning and the thematic desk review of the Malaria Programme Review (MPR). We will integrate the recommendations of the desk review and our modelling and analytics outputs into the development of the 2026-2030 National Malaria Strategic Plan. I conducted data cleaning on confirmed uncomplicated malaria cases among children under five years from the District Health Information System-2 (DHIS-2). I used the monthly case data to calibrate a vector-host compartmental model and predict SMC impact using the R and Python programming softwares.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2009 - 2010	House Officer, University of Nigeria Teaching Hospital, Enugu
2010 - 2011	Medical Research Officer, Health Policy Research Group, Enugu
2012 - 2019	Resident Physician, Department of Community Medicine, Federal Medical Centre Umuahia
2020 – 2024	Global Health Specialist Physician, Department of Community Medicine, Federal Medical Centre Umuahia
2024 -	Programme Associate, Health Division, Corona Management Systems, Abuja
2021 -	Senior Lecturer (Adjunct), Faculty of Public Health, University of America Centre, Enugu
2021 - 2022	National Technical Lead, Engaging Faith Communities for COVID-19 Surveillance and Risk Communication Project, Nigeria
2021 - 2022	Member, National COVID-19 Surveillance Pillar, National COVID-19 Incident Command Centre, Nigeria Centre for Disease Control, Abuja
2021 –	Member, Pandemic Preparedness Working Group, Junior Doctors Network, World Medical Association
2024 -	Mentor, Cohort 2, Forecasting for Social Good, AFRICAST Project
2024 -	Mentor, Cohort 2 & 3, Nigeria Malaria Modelling Fellowship, Corona Management Systems
2024 -	Assistant General Secretary, Applied Malaria Modelling Network (AMMnet), Nigeria Chapter
2024 –	Member, Learning Committee, AMMnet (Global)
2024 –	Reviewer, AMMnet Training Grants
2025 –	Reviewer, Abstracts for AMMnet 2025 Annual Meeting, Senegal
2025 -	Reviewer, American Journal of Tropical Medicine and Hygiene
2025 -	Member, Malaria Programme Review Committee, NMEP, Abuja
2025 –	Member, Malaria Indicator Survey Implementation Committee, NMEP, Abuja
2025	Instructor, Forecasting for Social Good, AFRICAST Project
2025	GBD (Global Burden of Disease) Collaborator

Honors

2002 - 2009	Undergraduate Scholar, Shell Petroleum Development Company Undergraduate Merit Award
2015	Best Candidate in Membership Examinations, Faculty of Public Health, National Postgraduate Medical College of Nigeria
2017 - 2018	Masters' Scholar, World Bank Graduate Scholarship Programme
2023	Malaria Modelling Fellow, Corona Management Systems / Gates Foundation
2024	Elizabeth Halloran Fellow, Emory Rollins School of Public Health, United States

2024 Visiting Scholar, Faculty Enrichment Program in Applied Malaria Modelling using Epidemiological Modelling (EMOD) software (Institute for Disease Modeling, Gates Foundation), University of Health and Allied Sciences, Ghana / Northwestern University, United States.
2024 Speaker, Mathematical Modelling Journal Club, Centre for Communicable Disease Dynamics, Harvard T. Chan School of Public Health, United States.
2025 Fellow, Epiverse Epidemiological Data Science Training for Africa, 2025
2025 AI Policy Fellow, TechImpact Forward, Canada

C. Additional Certifications

- **Institute for Health Metrics and Evaluation, 2025**
  Introduction to GBD online course
- **TechForward Impact, Canada, 2025**
  Foundations of AI Safety and Governance Bootcamp
- **University of Geneva, Switzerland Infectious Disease & Mathematical Modelling Division (May 2023)**
  Data Analysis with R, with Public Health Applications
- **Institute for Statistics and Data Science, Netherland (Sept. 2022)**
  R Training – Essential Level
- **University of Washington, USA (Sept. 2021)**
  Policy development and Advocacy for Global Health
- **Institute Pasteur de Tunis Regional Training Centre, Tunisia (May 2021)**
  implementation Research with a focus on Infectious diseases of poverty
- **University of Washington, USA (June. 2020)**
  Certificate in Fundamentals of Global Health Research
- **Collaborative Institutional Training Initiative, USA (Dec 2020)**
  Responsible Conduct of Research (RCR-Basic)

D. Contribution to Science

1. Artificial intelligence and machine learning in health and disease

We examined how diverse real time data types, including mobility patterns, social media activity, wearable sensor data, environmental signals and electronic health records, can be successfully combined with machine learning approaches to enhance early diagnosis, forecast illness trajectories, and optimise intervention options.

a. Lawal OP, **Okeh DU** et al. Integrating real-time data and machine learning in predicting infectious disease outbreaks. *AJMAB* 2025; 10(1): 147-163.

b. Lawal OP, **Okeh DU**, et al. Rethinking mathematical modeling in infectious disease epidemiology: Towards a more responsive and integrated future. *Discover Public Health* 2025 (Manuscript submitted).

2. Modelling to support subnational strategic planning of malaria interventions in Nigeria

We applied time-series analysis using autoregressive integrated moving average capturing seasonality (SARIMA), Holt-Winters exponential smoothing (ETS), basic recurrent neural network, long short-term memory and gated recurrent unit models to study a holoendemic urban population of children under 15 years in Ibadan, Oyo State, Nigeria. The dataset analyzed comprised clinical data collected prospectively from health facilities in Ibadan between 1996 and 2014. We forecasted malaria prevalence and temperature trends in Ibadan for the following three years. I curated the data and co-conducted the analysis using R and Python programming softwares.

a. **Okeh DU**, Olalekan T et al. Predicting malaria burden in Nigeria using AI and geospatial models. 2025 (Manuscript submitted).

b. **Okeh DU**, Seasonality and trend forecasting of malaria prevalence in Nigeria. 2024 (Poster presentation: Institute for Disease Modeling Symposium, Oct 1-2, Gates Foundation, Seattle, Washington, United States).

c. Langat AK, **Okeh DU**, Mutinda JK, Aguegboh NS, Ofori M, Baranon MD, Bekalo DB, Akilimali A, Mwalili SM, Amegah AK, Kandala NB. Forecasting malaria prevalence with the temperature variation in Ibadan City, Nigeria: Utilising malaria clinical data from 1996-2014. 2024 (Manuscript submitted to Computational and Mathematical Biophysics).

3. Seasonal malaria chemoprevention impact modelling

a. Richardson S, Steele A, Scott S, Ainsworth J, Eckert E, **Okeh DU**, et al. Global medium-term research priority setting for seasonal malaria chemoprevention: results from an eDelphi study. BMJ Glob Health 2025;10:e017550.

b. **Okeh DU.** Modelling the impact of seasonal malaria chemoprevention on malaria incidence in Nigeria. 2024 (Oral presentation: American Society of Tropical Medicine and Hygiene Conference, November 13-17, United States).
c. **Okeh, DU, et al**. Estimating the community effect of seasonal malaria chemoprevention on malaria transmission dynamics in Nigeria. 2024 (Poster presentation: Multilateral Initiative on Malaria (MIM) Society 8th Pan-African Conference, April 21-28, Kigali, Rwanda).